

Azure Bastion Deployment to Remove Public RDP Access

PROJECT TOOLS — IMPLEMENTED CONFIGURATION

The tools below were used to implement secure browser-based access to a private Azure Virtual Machine while removing the need for a public IP address on the VM.

Secure Access & Networking

Tool	Purpose in this project	Cost
Azure Bastion	Provides browser-based RDP access to the Windows Virtual Machine without assigning a public IP address to the VM.	Paid
Azure Virtual Network (VNet)	Provides the private network for the Virtual Machine and Azure Bastion.	Unpaid
Azure Subnets	Separates the Virtual Machine and Azure Bastion into dedicated subnets.	Unpaid
Network Security Group (NSG)	Controls network traffic and allows RDP traffic from the Azure Bastion subnet to the Virtual Machine.	Unpaid
Azure Virtual Machine	Provides the private Windows environment accessed through Azure Bastion.	Paid

Azure Configuration & Management

Tool	Purpose in this project	Cost
Azure Portal	Used to create and configure the Resource Group, VNet, subnets, VM, NSG, Bastion, and public IP resources.	Unpaid
Azure Resource Group	Organizes and manages all resources used in the project.	Unpaid
Azure Public IP	Used by the Azure Bastion host for external browser-based access; the Virtual Machine itself has no public IP.	Usage-based

Project Documentation

Tool	Purpose in this project	Cost
Git / GitHub	Used for version control and storing project documentation and configuration files.	Free tier
Draw.io	Used to create the Azure Bastion architecture diagram showing the administrator, Bastion, VNet, subnets, and private VM.	Free tier

Project Use Cases

- Provide administrators with **browser-based RDP access** to the Windows Virtual Machine.
- Keep the **Virtual Machine without a public IP address**.
- Use a dedicated **AzureBastionSubnet** for the Bastion host.
- Use **Network Security Group rules** to control RDP traffic.
- Demonstrate successful access to the private VM through **Azure Bastion**.

Key Project Considerations

- The Virtual Machine uses private IP address **10.0.2.4**.
- The VNet uses address space **10.0.0.0/16**.
- The Virtual Machine is deployed in **VMSubnet (10.0.2.0/24)**.
- Azure Bastion is deployed in **AzureBastionSubnet (10.0.1.0/26)**.
- The VM is accessed through the **Azure Bastion browser connection**.
- The VM does **not** have a public IP address.
- An NSG rule allows TCP **3389** from the Bastion subnet to the VM.
- Azure Bastion and the Virtual Machine can incur Azure usage charges while deployed.

Implemented Tools Summary

Azure Portal	Resource Group	VNet	Subnets
Virtual Machine	NSG	Azure Bastion	Public IP for Bastion
Git / GitHub	Draw.io		

Scope note: This updated version lists only the tools and configuration described as actually implemented in the project. It does not claim implementation of Microsoft Entra ID, Azure RBAC, Azure Policy, Bicep/ARM templates, Azure Monitor, Log Analytics, or Azure Resource Graph.